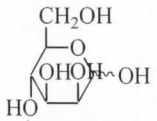
Organic Cumulative Examination
September 29, 2017

1) 3-O-methyl-D-mannose is found in mycobacterial polysaccharides and thus an important monosaccharide building block. Suggest a reasonable chemical synthesis toward this important target using D-mannose as starting material. (25 points)

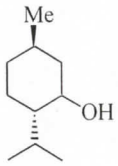


D-mannose

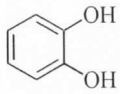
2) 3-O-methyl-D-mannose is a reducing monosaccharide. Give a definition for the term 'reducing monosaccharide', explain with chemical drawings what 'mutarotation' is, and describe in words what it measures (25 points)

3) Peracetylated α-glucopyranosyl bromide will yield almost exclusively peracetylated β-glucopyranosides upon reaction with an alcohol or thiol in the presence of a Lewis acid, such as BF_3 showing very high stereoselectivity for the reaction. Write out a reaction scheme for this transformation including key intermediates, and provide a rational for this observation (25 points)

4) The enzymatic pathways to synthesize glycosides suffer frequently from low yields in comparison to chemical procedures. Recent computational studies reflect the diverse influence of the aglycon during these transformations. Which reactivity and products can be expected during glycosylation of:
a) menthol
b) catechol
Give a detailed explanation for your prediction. (25 points)



menthol



catechol

5) Bonus question: Explain the term 'matrix effect' used in polymer chemistry and also in biochemistry. Elaborate on its nature, occurrence and/or use (10 points)